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OPERATING SYSTEM(ADVANCED)WK9

1. Scenario Analysis: Distributed Operating System in a Cloud Platform

A distributed operating system is used in a cloud platform to manage multiple connected computers and make them work together as one system. For example, a cloud service provider may use several servers to store files, host websites, and run applications for thousands of users. The distributed operating system allocates resources, manages processes, and allows communication between different computers. If one server fails, the system can transfer tasks to another available server to maintain service availability. It also supports load balancing by distributing workloads across servers to prevent overloading. Security features, such as user authentication and access control, help protect data from unauthorized access. Furthermore, distributed operating systems improve scalability because additional computers can be connected when demand increases. This makes cloud platforms more reliable, efficient, and flexible. Overall, a distributed operating system helps cloud providers deliver continuous services, manage resources effectively, and provide users with dependable access to applications and data.

2. Concept Research: Distributed Process Management

Distributed process management is the coordination and control of processes running on multiple computers connected through a network. It allows different systems to communicate, share resources, and execute tasks efficiently. In distributed computing environments, such as cloud platforms, process management handles task scheduling, process synchronization, communication, and workload distribution. It also monitors processes to detect failures and maintain system reliability. By distributing tasks across several computers, organizations can improve performance, scalability, and resource utilization. Distributed process management is widely used in cloud computing, distributed databases, and large enterprise systems. Overall, it ensures that multiple computers work together efficiently to complete complex tasks.

3. Tool Practice: Setting Up a Distributed System in Docker

Docker was used to understand how a distributed system can operate through multiple containers. The setup involved creating separate containers to represent different services connected through a shared Docker network. Each container performed a specific task and communicated with other containers when required. This practical exercise demonstrated how distributed applications can be organized, deployed, and managed efficiently. Docker made it easier to isolate services, configure environments, and start or stop components independently. It also showed how containers support scalability and portability. However, proper networking, security, and resource management are necessary. Overall, Docker simplifies the development, testing, and deployment of distributed systems.

4. Application Practice: Distributed System Architecture

A distributed system architecture consists of multiple interconnected computers or services that work together to perform tasks. The proposed architecture includes a client layer, a load balancer, application servers, a database server, and cloud storage. Users send requests through the client layer, while the load balancer distributes requests among application servers to prevent overloading. Application servers process requests and communicate with the database to retrieve or store information. Cloud storage keeps files and backups securely. If one server fails, another can continue providing services. This architecture improves scalability, reliability, performance, and resource sharing. It is suitable for e-commerce platforms, online banking, and cloud applications.

